Neural Network–Based Credit Risk Prediction Using a Multi■Layer Perceptron (MLP)

1. Dataset Description

This assignment focuses on predicting credit risk using a feedforward neural network model called a Multi■Layer Perceptron (MLP). The dataset used for this task contains financial and personal information of loan applicants. The objective of the model is to classify whether a person represents a credit risk or not. The dataset includes attributes such as credit duration, credit amount, installment rate, age, employment duration, housing status, savings information, and loan purpose. The target variable used for prediction is 'credit_risk'. The dataset was loaded using the Pandas library and explored to understand the structure and feature distribution.

2. Data Preprocessing

Before training the neural network, several preprocessing steps were applied to prepare the dataset. • Missing values were handled to ensure the dataset remained consistent. • Categorical features were converted into numerical format using one■hot encoding. • Numerical features were standardized using the StandardScaler technique. • The dataset was divided into training and testing sets using an 80■20 split. These preprocessing steps help improve model training and stability.

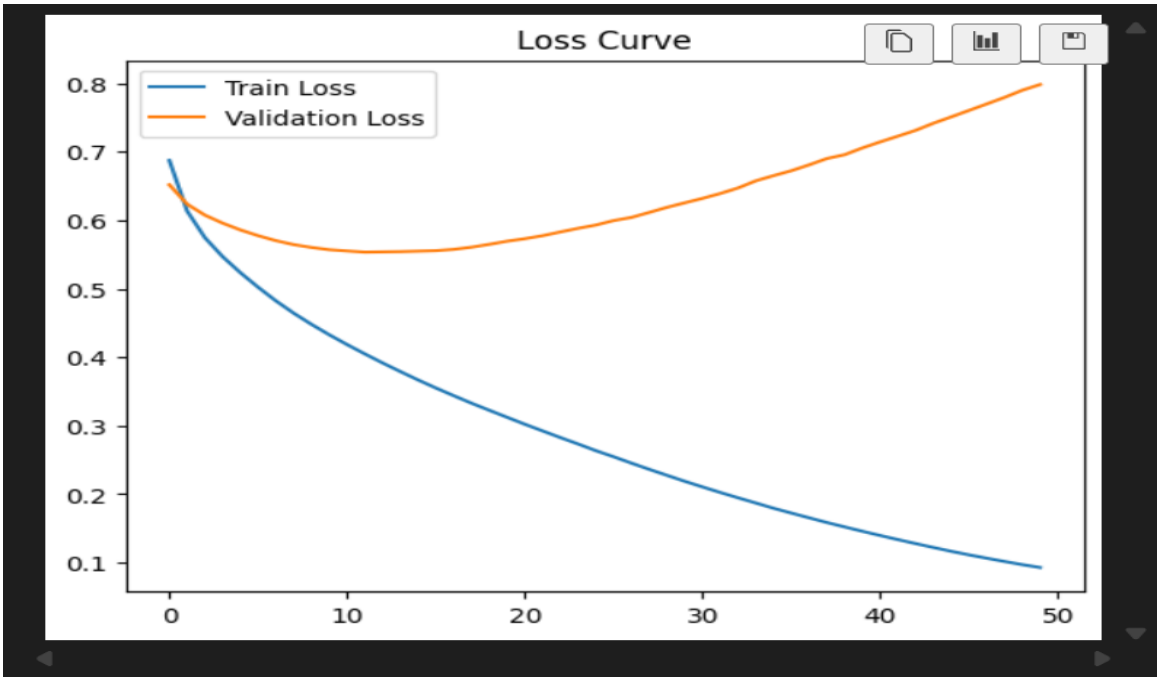
3. Model Architecture

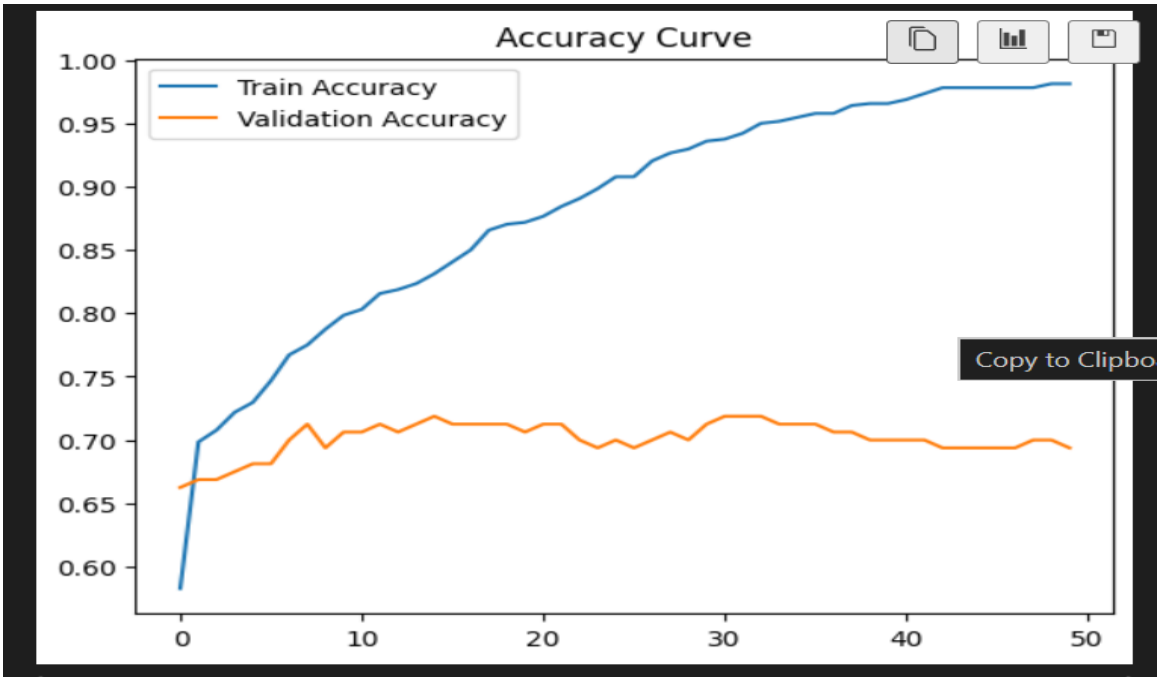
A Multi■Layer Perceptron neural network was implemented using the TensorFlow/Keras framework. The architecture of the model consists of: • Input Layer – receives all input features • Hidden Layer 1 – 32 neurons with ReLU activation • Hidden Layer 2 – 16 neurons with ReLU activation • Output Layer – 1 neuron with Sigmoid activation The model was compiled using Binary Cross Entropy loss and the Adam optimizer. Accuracy was used as the evaluation metric.

4. Training Results

The model was trained for 50 epochs. During training the model gradually improved its performance. Training accuracy increased from approximately 58% to about 98%. Validation accuracy stabilized around 70%, which indicates that the model successfully learned patterns from the dataset while still maintaining reasonable generalization.

Training Output and Model Logs


Loss Curve
Train Loss
Validation Loss
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50


Accuracy Curve
Train Accuracy
Validation Accuracy
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0
10
20
30
40
50
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7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
[[ 32  27]
 [ 24 117]]
              precision    recall  f1-score   support

           0       0.57      0.54      0.56        59
           1       0.81      0.83      0.82       141

    accuracy                           0.74       200
   macro avg       0.69      0.69      0.69       200
weighted avg       0.74      0.74      0.74       200
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Epoch 1/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 4s 38ms/step - accuracy: 0.5828 - loss: 0.6875 - val_accuracy: 0.6625 - val_loss: 0.6519
Epoch 2/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.6984 - loss: 0.6136 - val_accuracy: 0.6687 - val_loss: 0.6236
Epoch 3/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.7078 - loss: 0.5746 - val_accuracy: 0.6687 - val_loss: 0.6075
Epoch 4/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.7219 - loss: 0.5468 - val_accuracy: 0.6750 - val_loss: 0.5958
Epoch 5/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.7297 - loss: 0.5234 - val_accuracy: 0.6812 - val_loss: 0.5859
Epoch 6/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.7469 - loss: 0.5021 - val_accuracy: 0.6812 - val_loss: 0.5775
Epoch 7/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.7672 - loss: 0.4822 - val_accuracy: 0.7000 - val_loss: 0.5702
Epoch 8/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.7750 - loss: 0.4643 - val_accuracy: 0.7125 - val_loss: 0.5644
Epoch 9/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.7875 - loss: 0.4479 - val_accuracy: 0.6938 - val_loss: 0.5603
Epoch 10/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.7984 - loss: 0.4327 - val_accuracy: 0.7063 - val_loss: 0.5570
Epoch 11/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.8031 - loss: 0.4184 - val_accuracy: 0.7063 - val_loss: 0.5552
Epoch 12/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.8156 - loss: 0.4048 - val_accuracy: 0.7125 - val_loss: 0.5535
Epoch 13/50
...
Epoch 49/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9812 - loss: 0.0969 - val_accuracy: 0.7000 - val_loss: 0.7901
Epoch 50/50
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9812 - loss: 0.0928 - val_accuracy: 0.6938 - val_loss: 0.7984
```

5. Model Evaluation

The trained model was evaluated using several performance metrics including Accuracy, Precision, Recall, and F1■Score. A confusion matrix was also generated to analyze the classification performance. The results show that the neural network achieved an overall accuracy of approximately 74%, with strong recall and precision for the positive credit risk class. This demonstrates that the MLP model can effectively perform binary classification tasks for structured financial datasets.

Metric	Value
Accuracy	0.74
Precision (Class 1)	0.81
Recall (Class 1)	0.83
F1 Score (Class 1)	0.82

6. Conclusion

In this assignment, a Multi■Layer Perceptron neural network was successfully implemented to predict credit risk using tabular financial data. The experiment involved dataset preprocessing, neural network design, training, visualization of learning curves, and performance evaluation. The results demonstrate that neural networks can effectively model complex relationships in financial datasets and perform binary classification tasks such as credit risk prediction.